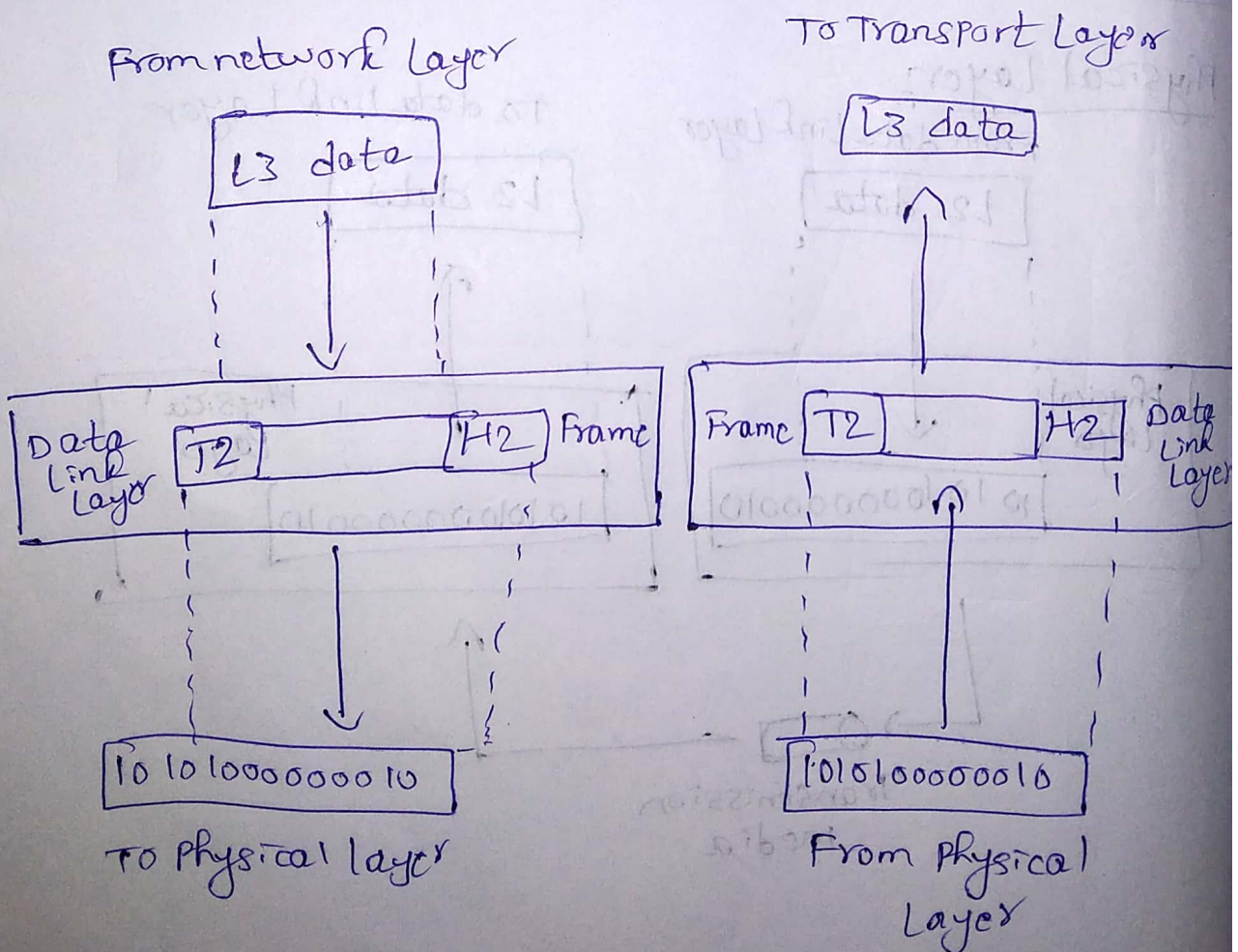


• Responsibilities of Physical Layer:

→ The physical layer is responsible for

- Data rate: The transmission rate - the no. of bits sent each second - is also defined by the physical layer
- Synchronization of bits: The send & receiver clocks must be synchronized
- Line configuration: It defines the way how two (or) more devices can be connected physically
- Physical topology: It defines the way how network devices are arranged
- Transmission mode: It defines the transmission mode whether it is simplex, half-duplex (or) full-duplex b/w the two devices on the network

• Data Link Layer:



• Data Link Layer

The data ~~layer~~ link layer is responsible for moving frames from one hop to the next.

• Responsibilities:

(1) Framing $\Rightarrow$ forming ^{Grouping individual bits to one bit.}

(2) Physical Addressing

~~(6) Logical Addressing~~

~~(7) Routing~~

(3) Flow control

(4) Error control

(5) Access control

• Flow control: If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.

• Error control: To detect and retransmit damaged or lost frames.

• Access control: When two or more devices are connected to the same link.

• Network Layer

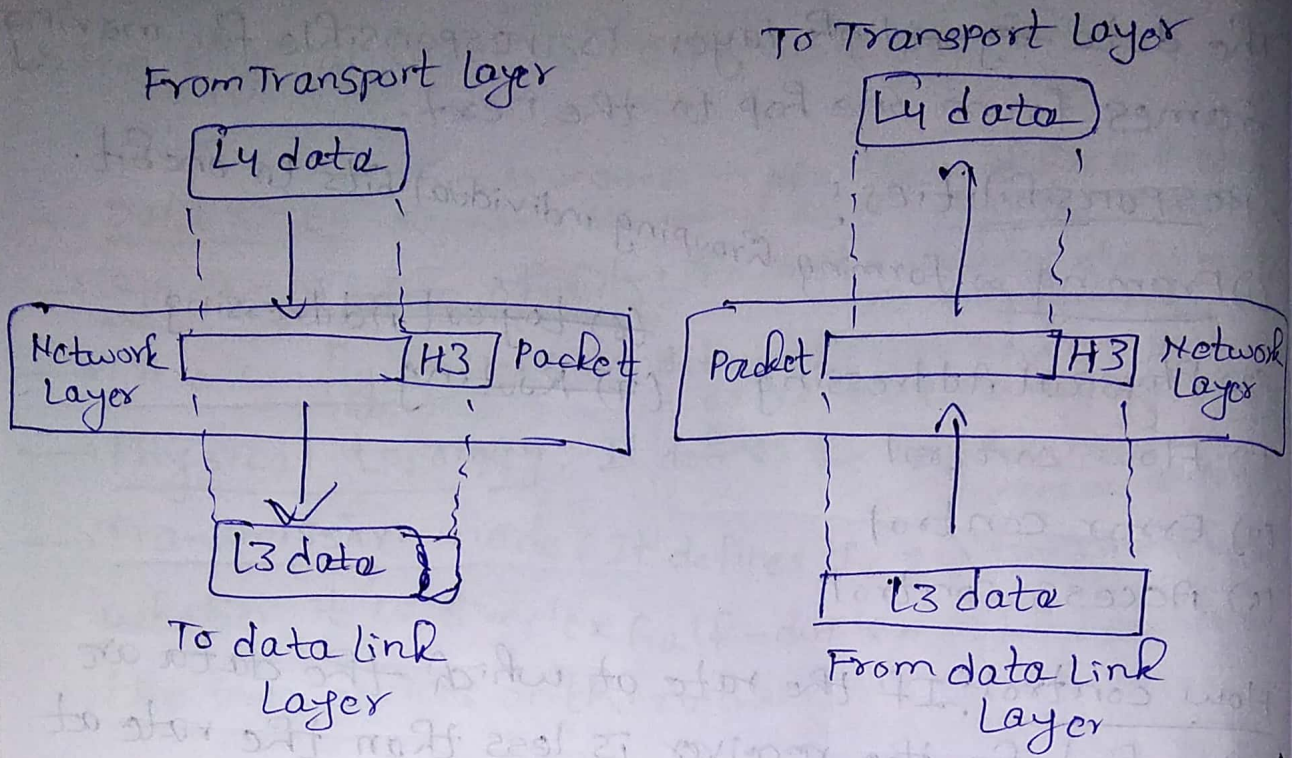
The network layer is responsible for the delivery of individual packets from the source host to the destination host.

• Responsibilities:

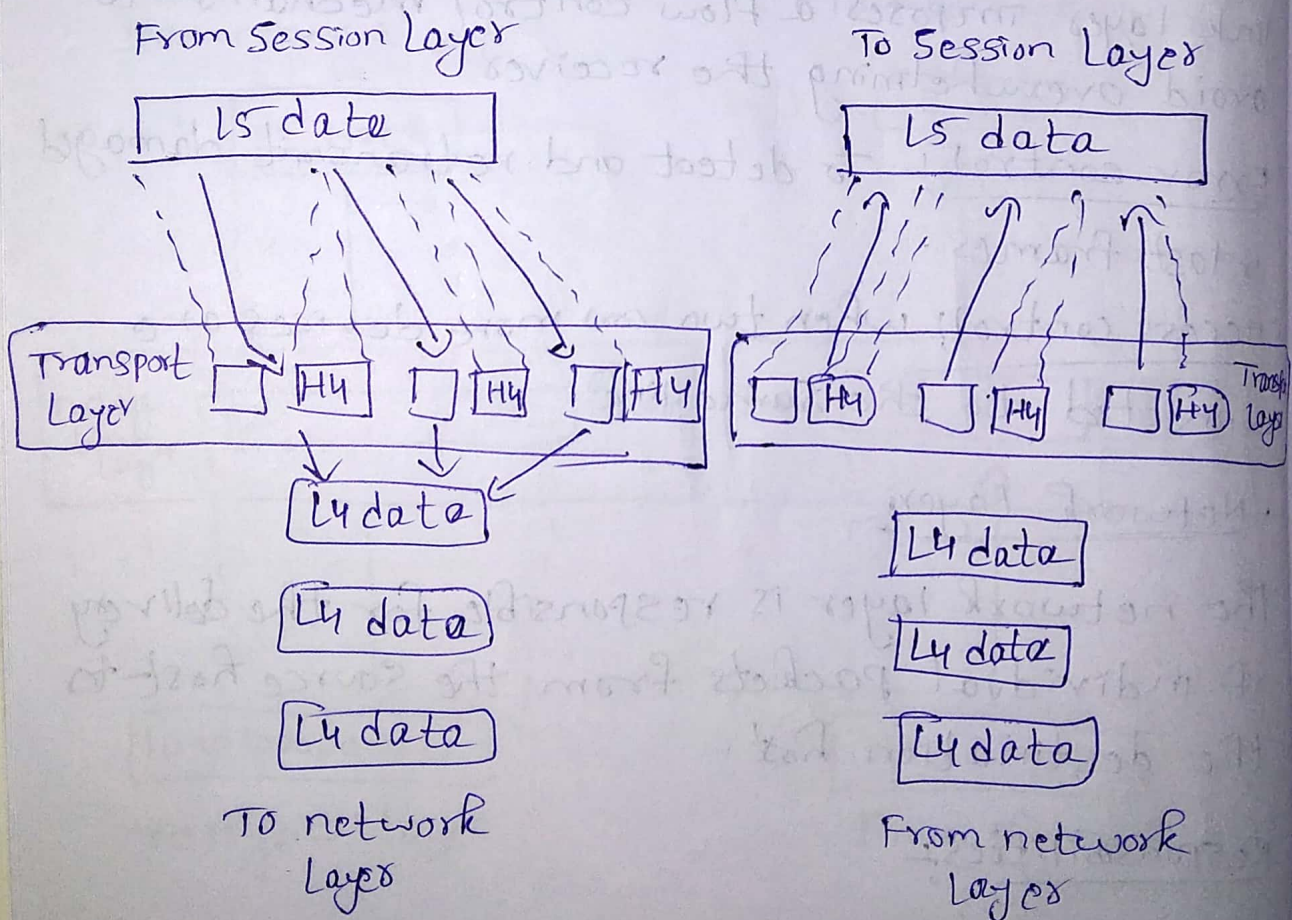
(1) Logical Addressing: handles the addressing problem locally

(2) Routing: when independent networks / links are connected to create internetworks (network of networks) or a large network the connecting devices (called routers).

* Network layer:



* Transport Layer:



Transport Layer:-

→ Transport layer is responsible for delivering a msg from one process to another.

→ The transport layer is a layer 4 ensures that msgs are transmitted in the order in which they are sent & there is no duplication of data.

→ Transport layer is responsible for transmitting the msg from one layer to another layer for this which contains protocols

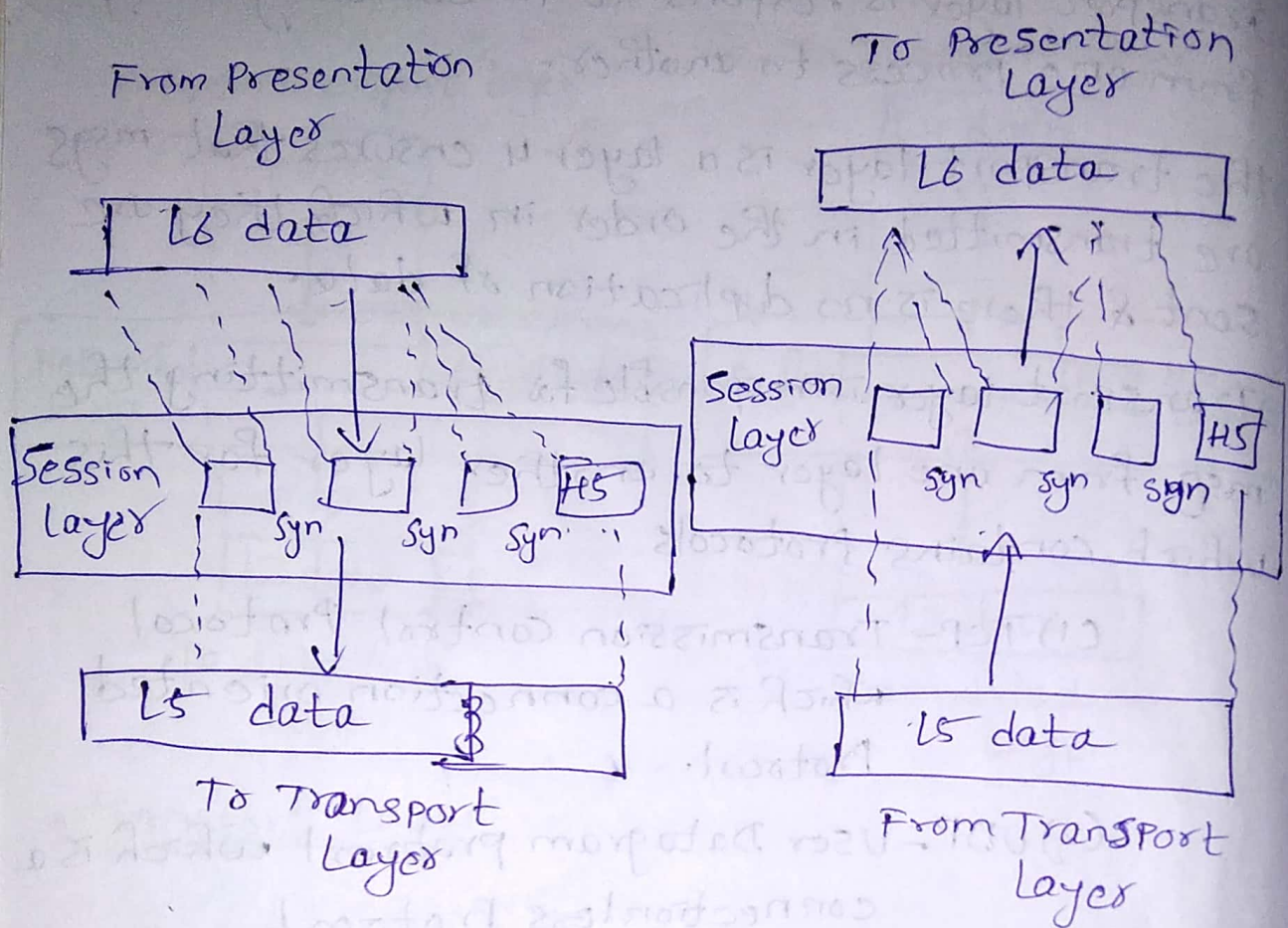
(1) TCP - Transmission Control Protocol which is a connection oriented Protocol.

(2) UDP - User Datagram protocol which is a connectionless Protocol

* Responsibilities!

- (1)
- (2) Segmentation & reassembly
- (3) connection control
- (4) Flow control
- (5) Error control
- (6) Access control

* Session layer:



Session layer:

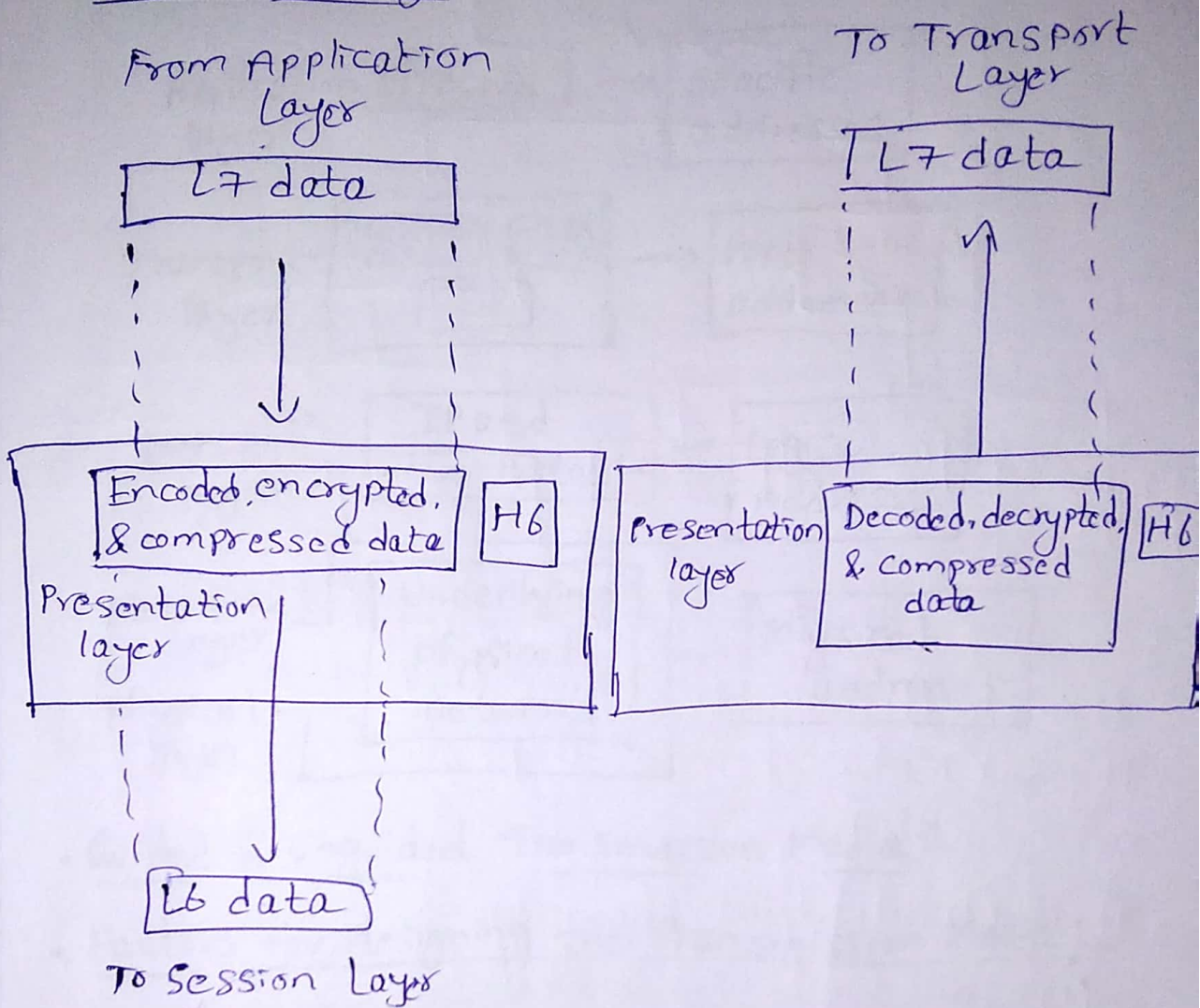
The session layer is responsible for dialog control & synchronization. The session layer is used to establish, maintain & synchronizes the interaction b/w communication devices

* Responsibilities:-

- (1) Dialog control
- (2) Synchronization

(4) Dialog control: The session layer allows two systems

• Presentation Layer:



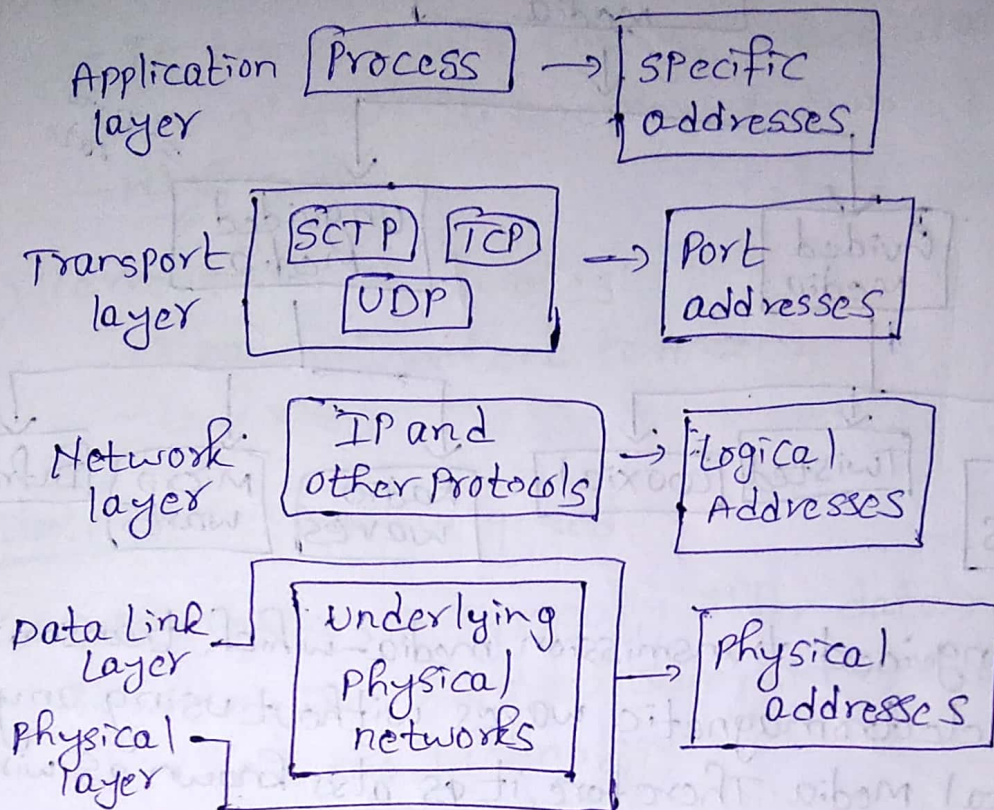
Presentation layer:

The Presentation layer is responsible for

• Responsibilities:

(1)

TCP/IP Model (Transmission control Protocol/Internet Protocol):



* Guided & unguided Transmission Media:-

* Factors for designing the Transmission Media:-

(1) Bandwidth

(2) Transmission Impairment

(3) Interference $\Rightarrow$ Interrupting the services

* causes of Transmission Impairment

(1) Attenuation $\Rightarrow$ which means loss of energy i.e. the strength of signal decreases with increase the distance

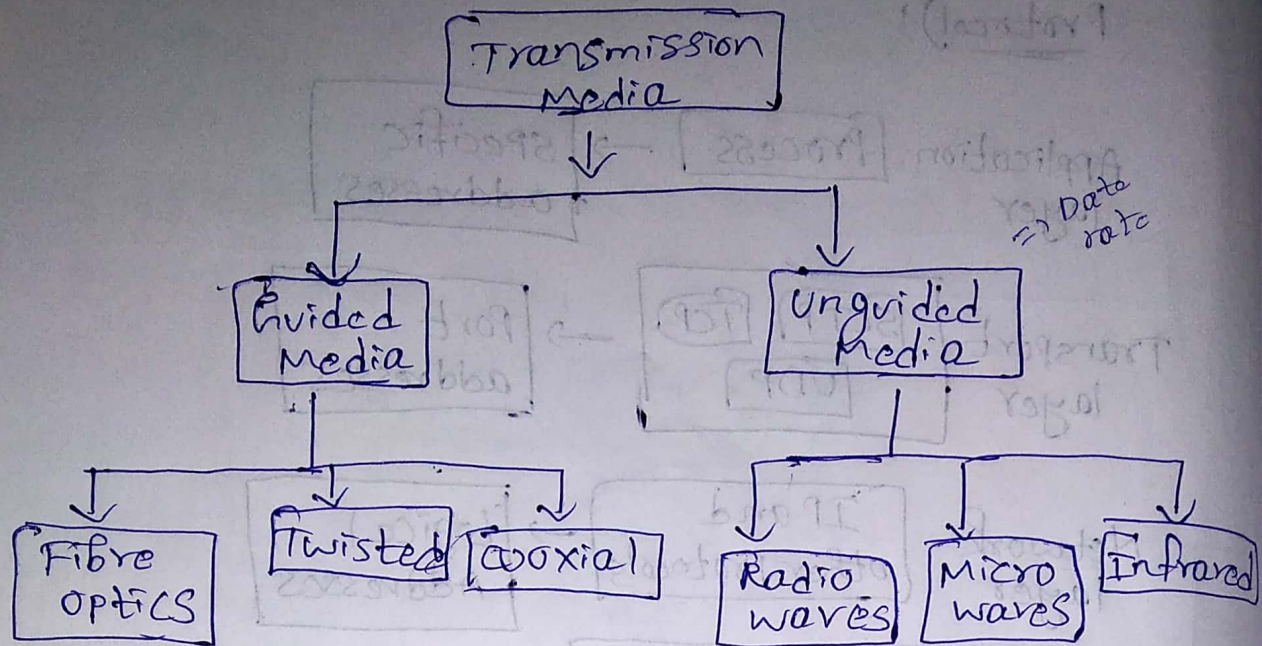
(2) Distortion $\Rightarrow$ change in shape of the signal.

(3) Noise $\Rightarrow$ Due to some extra signals we may cause the noise. (unwanted signals)

* Transmission media:-

It is a media that carries information from sender to receiver.

* classification of Transmission Media



→ In unguided transmission media - which transmits the electromagnetic waves without using any physical media. Therefore, it is also known as wireless communication.

→ It is classified into three types -

- (i) Radio waves ~~→ It is only for~~
- (ii) Micro waves
- (iii) Infrared

(i) Radio waves!

Applications!

→ It can penetrate signals to Building

→ Local Area development possible

→ The radiowaves are useful for multicasting

→ FM Radio, Television, cordless communication

(ii) Micro waves:

→ Micro waves can transmit two ways —

b) ^{Satellite} ~~terrestrial~~ microwaves

- It transmit data through space station
- one station to another station
- High cost

a) Terrestrial microwaves

- Transmit Antenna to Antenna
- Low rate

* Data Rates: of unguided:

(i) Radio waves : Typically offer data rates from a few kbps to several Mbps, depending on the technology.

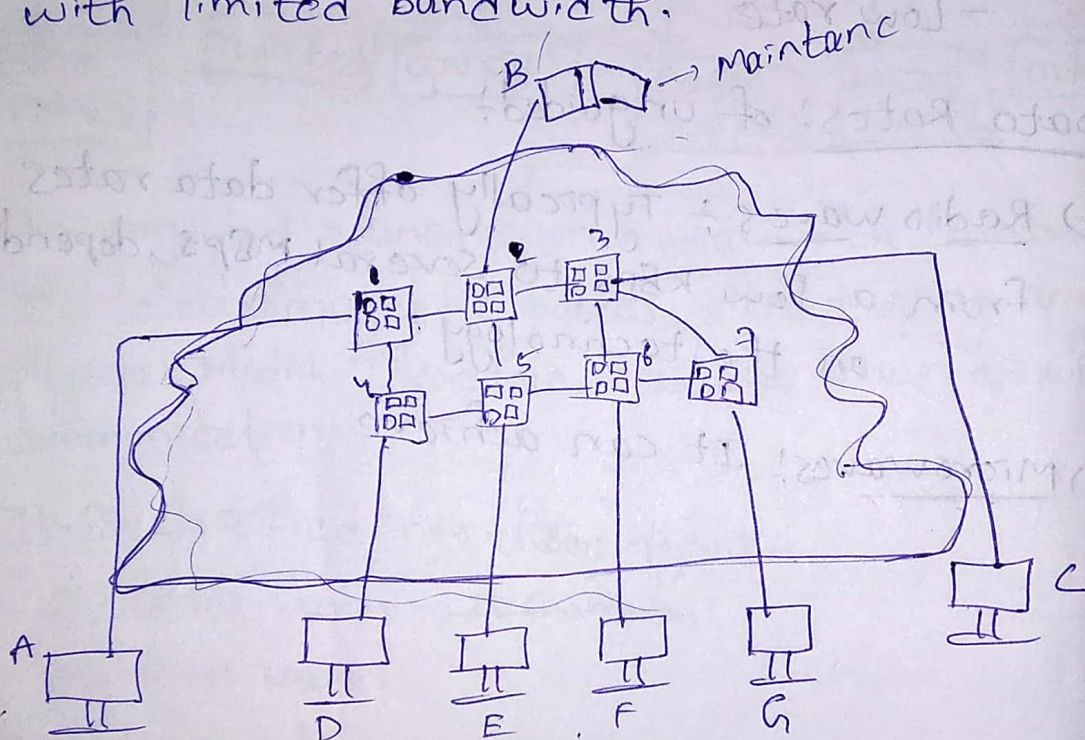
(ii) Microwaves: It can achieve

* Switching ÷ switching is a technic transferring information from one computer network to another network.

→ switching in a computer network is achieved by using switches.

→ It is operated in full duplex mode.

→ It does not broadcast the msg as it works with limited bandwidth.



* Simple switching ÷

→ Three types of switching techniques

(1) circuit switching

(a) space division switching

(b) Time division switching

(2) Message switching

(3) Packet switching

(a) Datagram packet switching (b) virtual circuit Packet switching

(1) circuit switching :-

circuit switching is a switching technique that establishes a dedicated path b/w sender and receiver.

* Places in circuit switching :-

(1) connection establishment

(2) Data Transfer

(3) connection teardown / Termination

(2) Message switching :-

message switching is a switching technique in which msg is transferred as a complete ^{unit} and routed through intermediate as it stores

* Design Issue of Data Link Layer :-

→ The Data Link layer in the OSI model, is in b/w the physical layer & the network layer. This layer converts the raw transmission facility provided by the physical layer to a reliable and error free link.

→ The main function and the design issues of this layer are -

(1) Providing services to the network layer.

(2) Framing

(3) Error control

(4) Flow control

→ In the OSI model, the data link layer uses the services offered by the physical layer. The primary function of this layer is to provide a well-defined service interface to network layer above it.

→ The types of services provided can be of three types -

(i) unacknowledged ~~connectionless~~ connectionless service

(ii) Acknowledged connectionless service

(iii) Acknowledged connection-oriented service.

* Framing:-

* Error Detection Methods:-

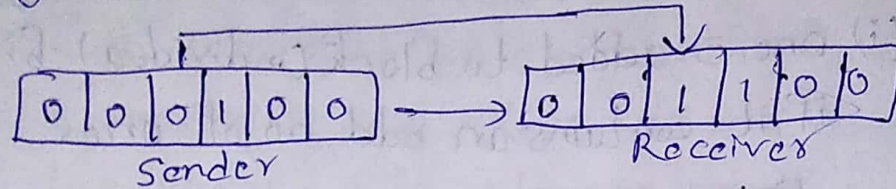
→ Error is a condition when the receiver's information does not match the sender's.

→ Digital signals suffer from noise during transmission that can introduce errors in the binary bits travelling from sender to receiver. That means a zero bit may change to one (or) a one bit may change to zero.

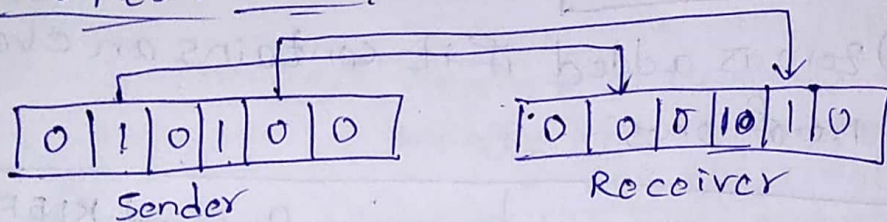
* Types of Errors:

- (1) Single Bit Error
- (2) Multiple Bit Error
- (3) Burst Error.

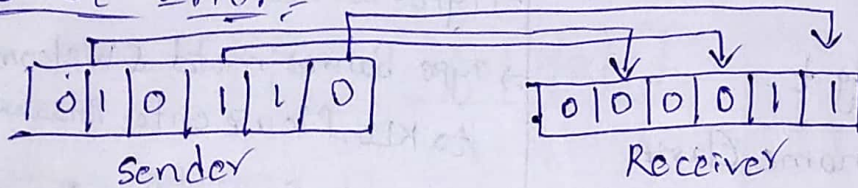
(1) Single Bit Error:



(2) Multiple Bit Error:



(3) Burst Error:



* Error Detection Methods:

→ To detect errors a common technique is to introduce redundancy bits that provide additional information. ~~where~~ various

- (1) Simple parity check
- (2) Two dimensional Parity check
- (3) Checksum
- (4) cyclic redundancy check (CRC)

(1) Simple Paraty check:-

→ Simple Bit paraty is a simple error detection method that involves adding an extra bit to a data transmission

→ It works as -

combination
of 0's and 1's

(i) one is added to block (individual bits) if it contains an odd no. of ones

1	0	1	0	1
---	---	---	---	---

 ⇒ There are three ones

(ii) Zero is added if it contains an even no. of ones

⇒ Type en → To enable
After

⇒ Type config t

⇒ Type hostname ClassA

⇒ Type line console 0

↳ we have entered the console

⇒ Password KLV123

⇒ Type login

⇒ Type exit

⇒ Type line vty 0 15

⇒ Type Password KLV123

⇒ Type login

⇒ Type exit

⇒ Type enable Secret KLEF12

⇒ Type service Password-encryption

⇒ Type banner motd & Welcome
to KLV, Please enter PASSWORD &

⇒ Type interface vlan 1

⇒ Type IP address 128.107.20.10
256.256.256.0

⇒ Type no shutdown

⇒ Type exit

⇒ Again Type exit

⇒ Type

motd ⇒ msg of the day

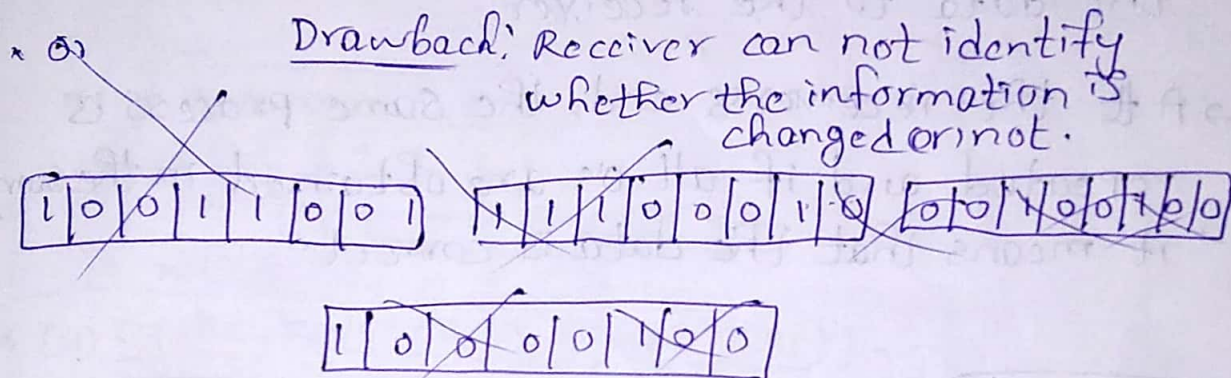
vty ⇒ virtual terminal

t ⇒ terminal

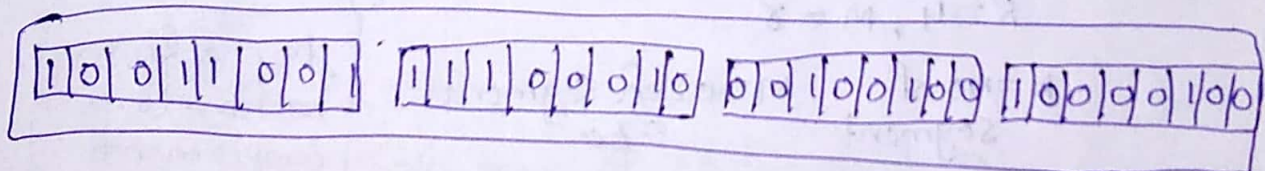
(2) Two dimension Paraty check

→ Two dimensional Paraty checks are calculated each row which is equivalent to simple Paraty check bit.

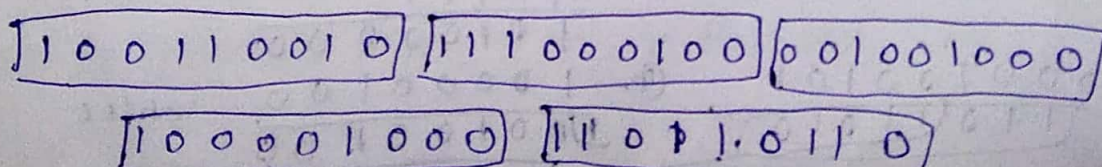
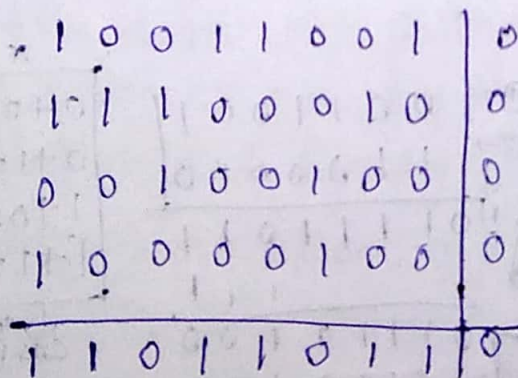
→ Here bits are also calculated for all columns, then both are sent along with the data. At the receiving end these bits are compared with the paraty bits calculated on the received data.



• original data:



• original data:



(3) checksum!

→ checksum error detection is a method used to identify errors in transmitted data.

→ The process involves dividing the data into equally sized segment and using a ones complement to calculate the sum of these segments.

→ The calculated sum is then sent along with the data to the receiver.

→ At the receiver's end, the same process is repeated, and if all 0's are obtained in the sum, it means that the data is correct.

10011001 11100010 00000100 10000100

$K=4, M=8$

$K = \text{no. of segment}$

$M = \text{each segment size}$

In checksum, we have to do complement i.e. 0 is replaced to 1

* Sender side:

10011001
11100010
00000100
10000100

$\begin{array}{l} 0+0=0 \\ 0+1=1 \\ 1+0=1 \\ 1+1=0 \end{array}$

Carry=1

01111100
00100100
10100000

10000100
000100100

Sum → 00100101

checksum

ctrl+c

* Receiver side:

① → 1 0 0 1 1 0 0 1

② → 1 1 1 0 0 0 1 0

1 0 1 1 1 0 1 1
 1 0 1 1 1 0 1 1

0 1 1 1 1 0 0

③ → 0 0 1 0 0 1 0 0

1 1 1 1 1 1 1 1

1 0 0 0 0 0 0 0

④ → 1 0 0 0 0 1 0 0

1 0 0 0 0 0 1 0 0

0 0 0 0 0 1 0 1

check sum → 1 1 0 1 1 0 1 0

1 1 1 1 1 1 1 1

⇒ 0 0 0 0 0 0 0 0

⇒ (Sum)

ctrl+v

complete

⇒ complement the sum we get All zeroes.

So, the receiver data is accepted

* (u) cyclic redundancy check (CRC):

unlike the checksum scheme, which is based on addition, CRC is based on Binary division.

Sender side:

In CRC, a sequence of redundant bits, called cyclic redundancy bits are appended to the end of the data unit, so that the resultant data unit becomes exactly divisible by a second, predetermined binary number.

At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is there for accepted

* CRC working:

we have given data word of length n and divisor of k

Step-1: Append $k-1$ 0's to the original msg

Step-2: Perform module of two division

Step-3: Remainder of division is equal to CRC

Step-4: code word is equals to data width append $k-1$ 0's + CRC

NOTE: CRC must be $k-1$ bits

NOTE: Length of code word = $n + k - 1$ bits

Example: Let's data to be sent is

1 0 1 0 0 0 0 and divisor in the form of Polynomial is $x^3 + 1$

Sol:

$K = 1001$

So, $K = 4 \Rightarrow K - 1 = 4 - 1 = 3$

$x^3 + 1 \rightarrow x^3 + x^2 + x + 1$
1 0 0 1

So, add 3 zeroes

1001 | 1010000000

1001

0011000000

1001

01010000

1001

01010

1001

011

1010000000

011

1010000011

forwarded to receiver side

Receiver Side

1010000011

100 | 1010000011
1001

00 | 11000011
1001

0 | 1010011
1001

00 | 10011
1001

0 | 1001
1001

0000

* Error correction!

$$2^r \geq m + r + 1$$

$$m = 1010 \Rightarrow 4$$

disadvantages

It supports only for single-bit

$$(i) r=1 \Rightarrow 2^1 + 1 + 1$$

$$= 6 \times$$

$$2 \geq 6$$

$$(ii) r=2 \Rightarrow 2^2 = 4 \Rightarrow 4 + 2 + 1 = 7$$

$$4 \geq 7 \times$$

$$(iii) r=3 \Rightarrow 2^3 = 8 \Rightarrow 4 + 3 + 1 = 8$$

$$8 \geq 8 \checkmark$$

$r \rightarrow$ redundant bit

$n \rightarrow$ size

$m \rightarrow$ original size

TCP $\rightarrow$ connection oriented

UDP $\rightarrow$ connection less

$$\begin{array}{ccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & P_3 & 0 & P_2 & P_1 \end{array}$$

$$3 \Rightarrow 2^3 = 8$$

$$P_1 \rightarrow 1, 3, 5, 7 \rightarrow P_1 \begin{array}{ccc} 0 & 1 & 1 \\ 0 & 0 & 1 \end{array} \rightarrow 0$$

$$P_2 \rightarrow 2, 3, 6, 7 \rightarrow P_2 \begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \rightarrow 1$$

$$P_3 \rightarrow 4, 5, 6, 7 \rightarrow P_3 \begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 0 \end{array} \rightarrow 0$$

$$\begin{array}{ccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{array}$$

$$P_1 \rightarrow 1, 3, 5, 7 \rightarrow 0011 \rightarrow \underline{0001} \rightarrow 1$$

$$P_2 \rightarrow 2, 3, 6, 7 \rightarrow 1001 \rightarrow \underline{1001} \rightarrow 0$$

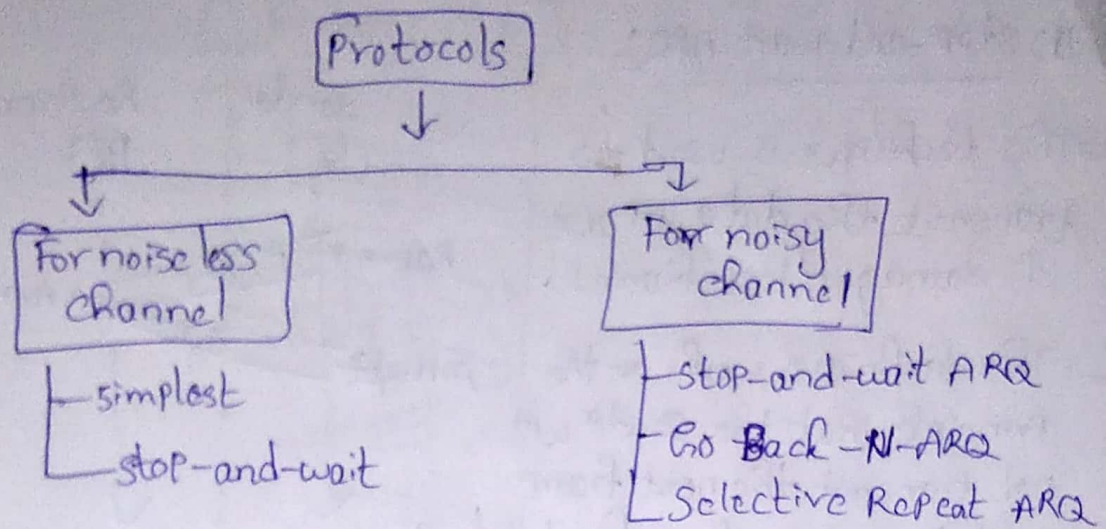
$$P_3 \rightarrow 4, 5, 6, 7 \rightarrow 0101 \rightarrow \underline{0001} \rightarrow 1$$

So, it is

$$101 \Rightarrow 5$$

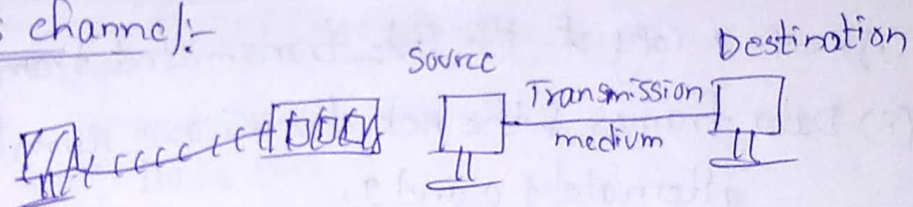
$$\begin{array}{ccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \end{array}$$

* Data Link Layer Protocols:-



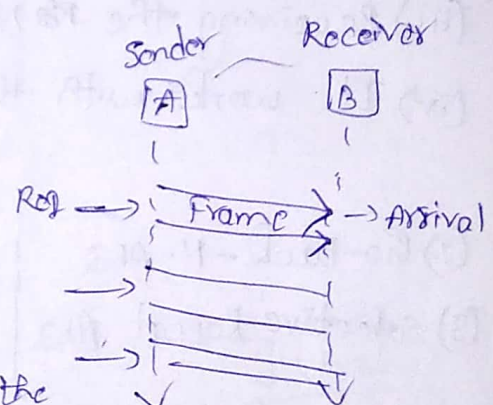
→ ARQ = Automatic Repeat Request

* Noiseless channel:-



(1) Simplest?

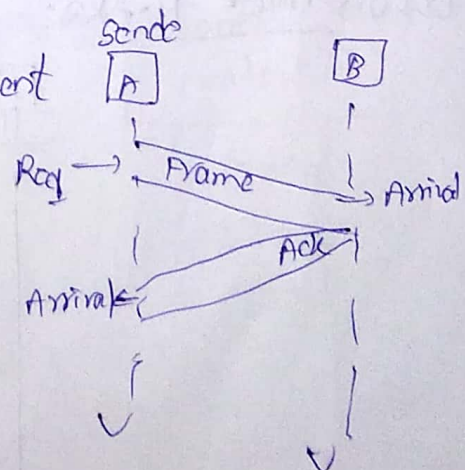
The sender sends



(2) stop-and-wait?

The sender will not transmit the next frame

There is a concept of acknowledgement of each frame.

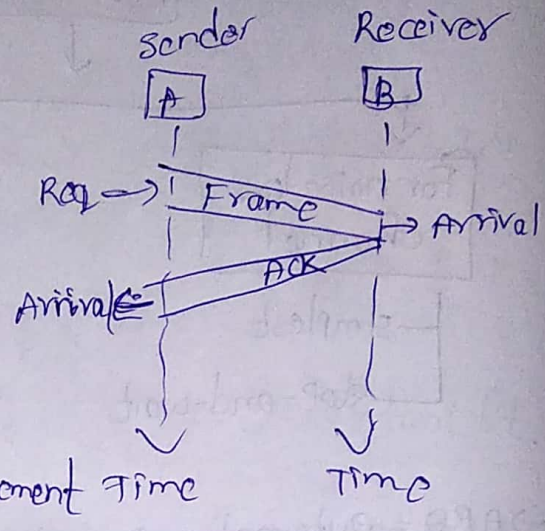


* Noisy channel

(1) Stop-and-wait ARQ

→ This technique is used to transmit the data in case of damaged/lost frames

→ This technique works on the principle that the sender will not transmit the next frame until it receives the acknowledgement time



* Four features are required for the transmission

(i) keep a copy of the last transmitted frame

(ii) Data frames & the ACK frames are numbered alternately 0 and 1.

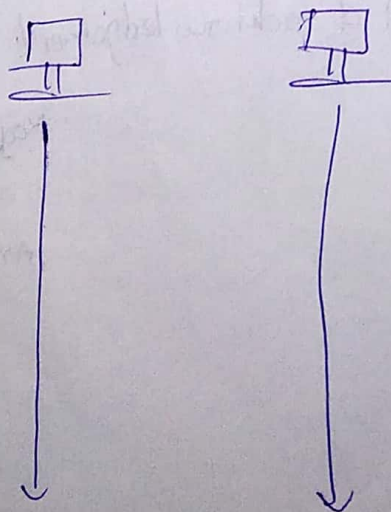
(iii) Receiving the ~~no~~ NAK frame

(iv) It works with the timer.

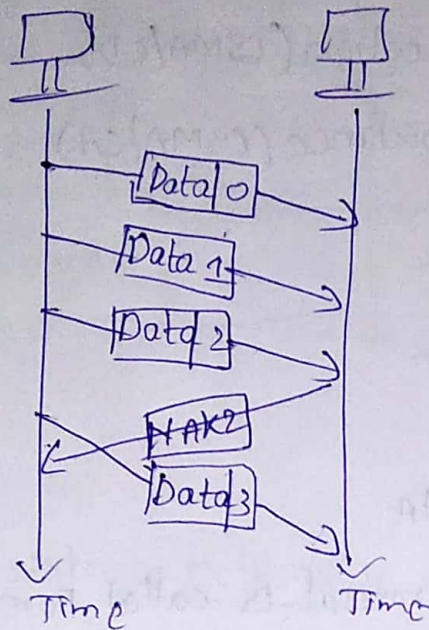
1 frame = group of bits
1 sliding = group of frames

(2) Go-Back-N-ARQ } These are sliding windows
(3) selective Repeat ARQ }

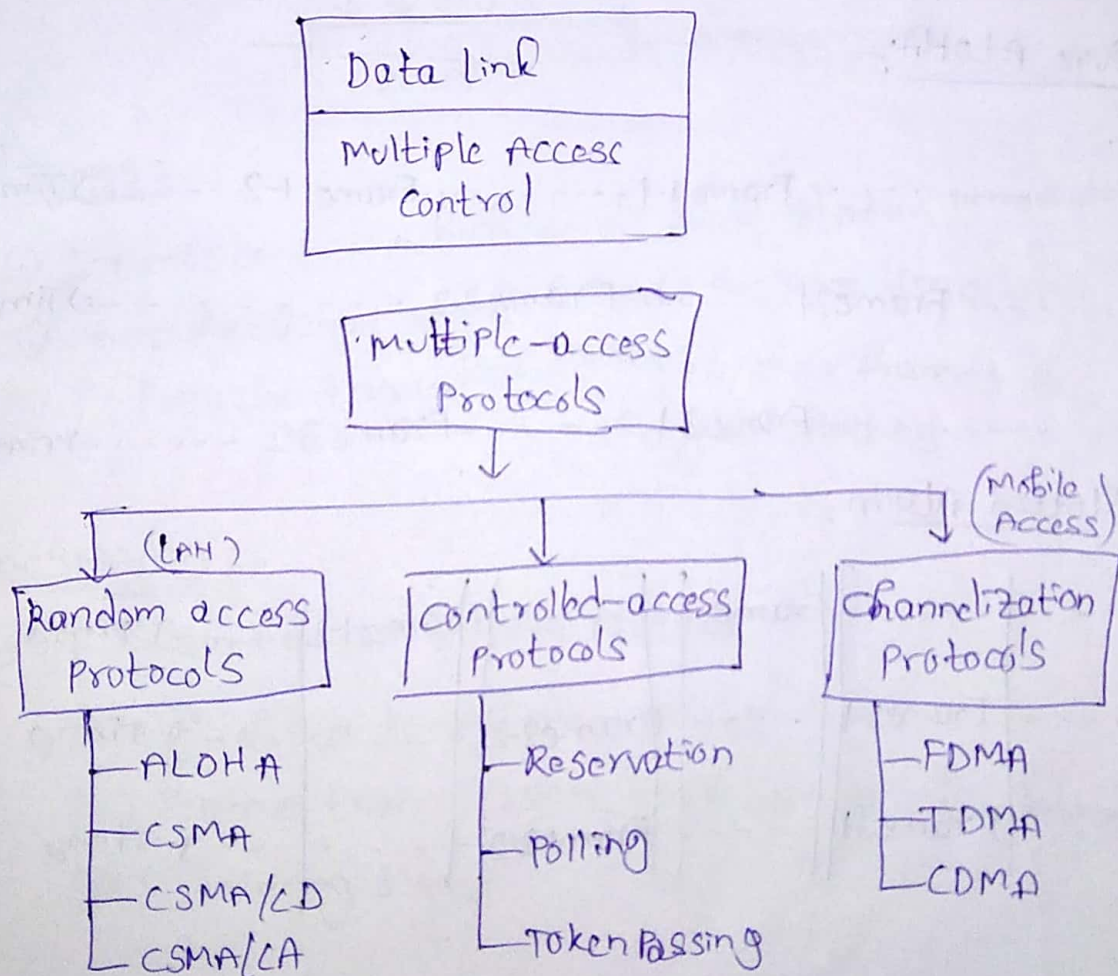
(2) Go-Back-N-ARQ:



(3) Selective Repeat ARQ



x Medium Access Control (MAC) Protocols



- Carrier Sense Multiple Access (CSMA)
- CSMA/Collision Detection (CSMA/CD)
- CSMA/Collision Avoidance (CSMA/CA)

(i) ALOHA Protocols :

→ There are two types -

(i) Pure ALOHA

(ii) Slotted ALOHA

→ The original ALOHA Protocol is called Pure ALOHA.

→ The idea is that each station sends a frame whenever it has a frame to send (multiple access)

Pure ALOHA:

----- Frame 1-1 ----- Frame 1-2 -----> Time

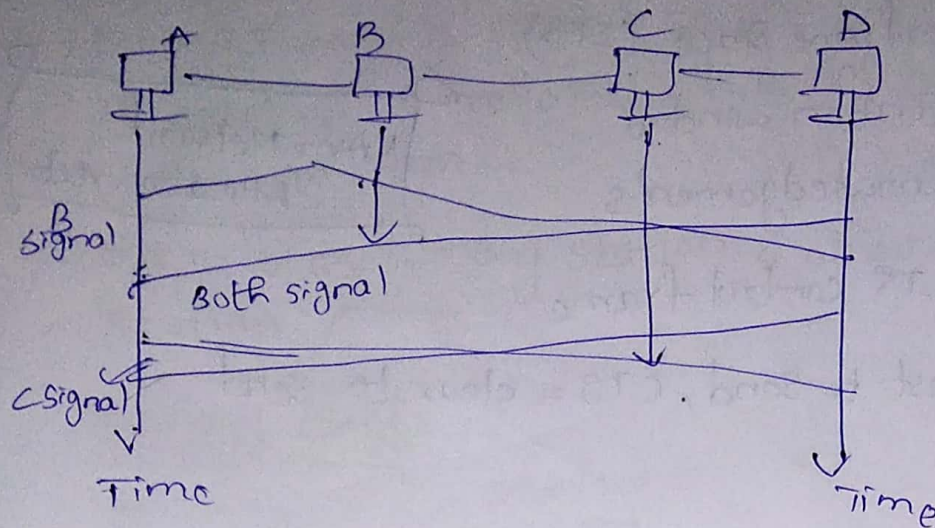
----- Frame 2-1 ----- Frame 2-2 -----> Time

----- Frame 3-1 ----- Frame 3-2 -----> Time

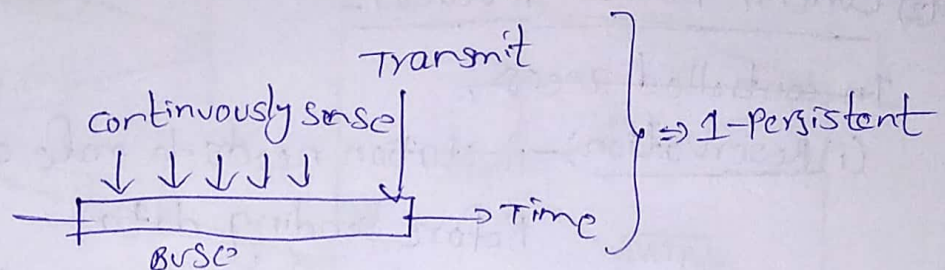
Slotted ALOHA:

-----	Frame 1-1	-----	Frame 1-2	-----> Time
-----	Frame 2-1	-----	Frame 2-2	-----> Time
-----	Frame 3-1	-----	Frame 3-2	-----> Time

(2) CSMA:



CSMA Persistence Methods :-



Types :-

- (i) 1-Persistent $\Rightarrow$ which sends each station immediately
- (ii) Non-persistent $\Rightarrow$ which waits for some time
- (iii) P-Persistent $\Rightarrow$ which transmit their frames to the given slotted

(3) CSMA/CD :-

$\rightarrow$ It is similar to ALOHA Protocol

- (i) Addn of the persistence process
- (ii) Frame transmission, wait for an acknowledgement
- (iii) Jamming signal

→ collision are avoided using three strategies

(i) Interframe Space (IFS)

(ii) Contention window

(iii) Acknowledgements

NAR = Network
Allocation vector

→ RTS and CTS control frame

RTS = Request to send, CTS = clear to send

~~→ (i) IFS~~

~~→ The co~~

* (2) Control Access Protocol :-

~~In controlled access,~~

(i) Reservation :- A station needs to make a reservation before sending data.

— only one device is allowed to transfer at a time.

* Advantages :-

→ High rates & low rates of data accessing time of respective channel can be predicated easily.

→ Provide predictable network performance

→ Reduce contention for network resources

Disadvantages :-

→ Control dependability

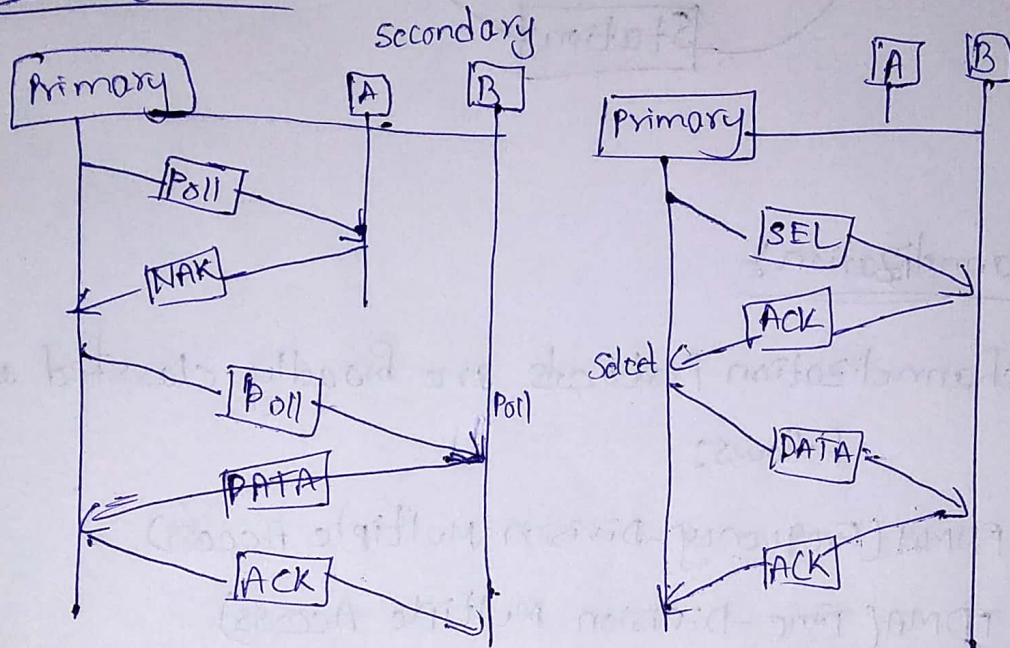
→ Decrease in capacity and channel data rate under light loads, Increase in turn-around time

(ii) Polling:

→ Polling Process is similar to the roll-call performed in class. A controller sends a msg to each node in turn.

→ one acts as Primary station & other secondary station.

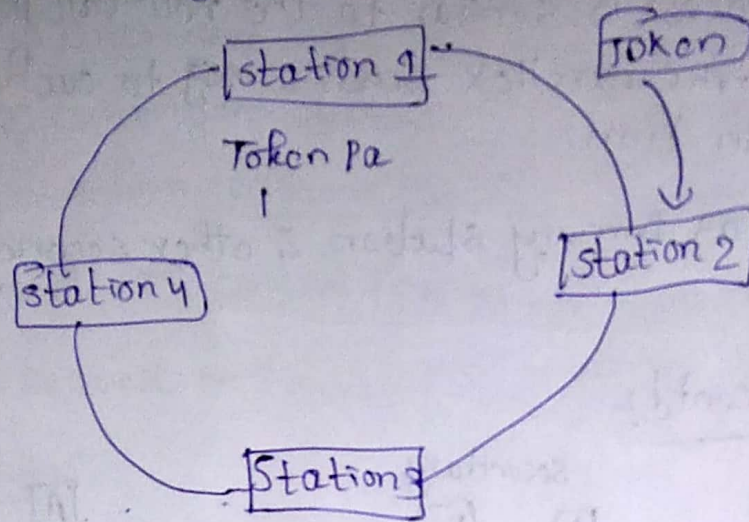
Primary Contd.:



Disadvantages:

- More time
- out of data to send
- Increase in the turnaround time leads to a drop in the data rates of the channel under low loads
- Efficiency = $Tt / (Tt + T_{Poll})$

(iii) Token Passing



* Channelization

→ Channelization Protocols are broadly classified as follows:

- FDMA (Frequency-Division Multiple Access)
- TDMA (Time-Division Multiple Access)
- CDMA (Code-Division Multiple Access)

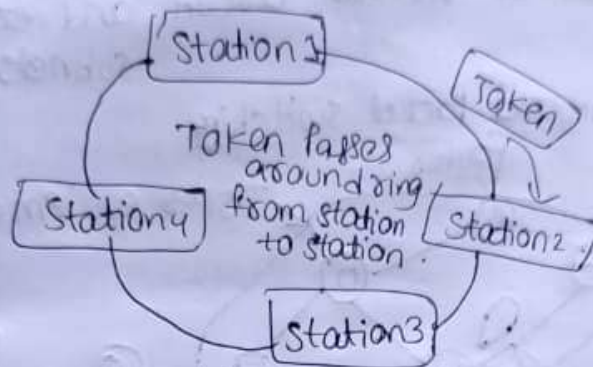
(i) FDMA

→ Each station makes use of the bandpass filter in order to confine the frequencies

Disadvantages:

- * Consume more time
 - * Efficiency = $T_t / (T_t + T_{Poll})$
- Let T_{Poll} be time for polling, T_t be the time required for transmission of data.

Token Passing:



Channelization:

- * Channelization Protocols
 - 1) FDMA (Frequency-Division Multiple Access)
 - 2) TDMA (Time-Division Multiple Access)
 - 3) CDMA (Code-Division Multiple Access).
- * A method that provides multiple access and available bandwidth of link is shared in time, frequency

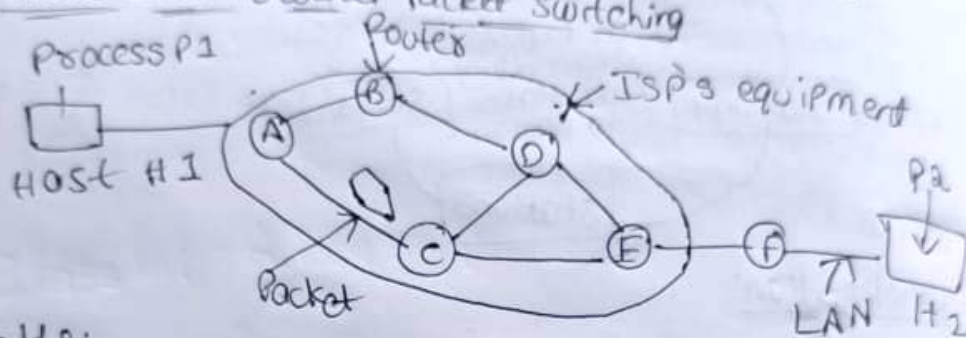
FDMA:

Time-Division Multiple Access

Network Layer Design Issues

- Store and forward Packet switching
- Services Provided to Transport layer.
- Implementation of Connectionless/Connection-oriented service and datagram subnets.

Store and Forward Packet Switching



Paths:

- H2 A → B → D → E → F H2
- H1 A → C → E → F H2
- H2 A → B → D → C → E → F H2

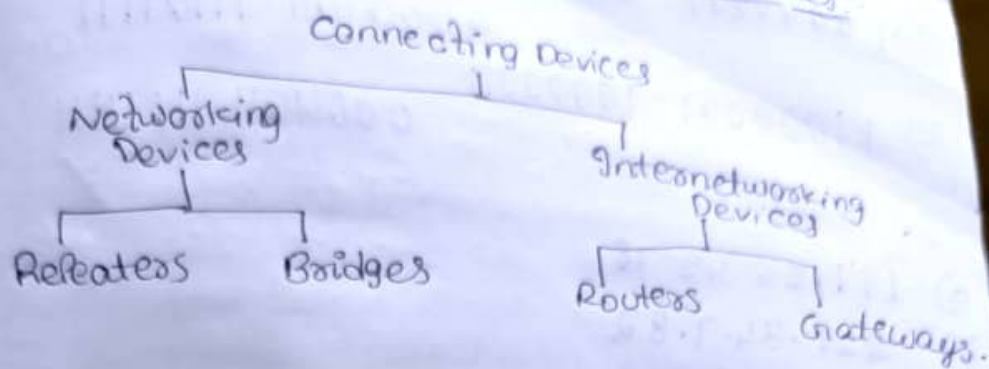
→ Connection oriented & Connectionless :-

* If connectionless service is offered, packets are injected into network individually & routed independently each other.

The packets are frequently called datagrams.

If connection-oriented is used a path from source router all the way to the destination router must be established before any data packets can be sent.

Networking and Internetworking Devices



IP Addressing:

- * IP addresses
- * IPv4 address
- * Address space
- * Notation
- * IPv6
- * Casting.

IPv4 Header

version	HL	type of service	Total length
Identification			Flags Fragment offset
Time to Live	Protocol	Header Checksum	
Source Address			
Destination Address			
options			padding

2³²

IPv6 Header

version	Traffic class	Flow label
Payload length	Next Header	HOP Limit
Source Address		
Destination Address		

128
2

a) 10000001 00001011

b) 11000001 10000011

00001011 11101111

00011011 11111111

a) 141.56.45.78

b) 221.34.7.82

Classful Addressing

→ binary notation

→ decimal-dotted notation

a) Binary notation

	First byte	second byte	third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

Drawback: Assigning IP addresses

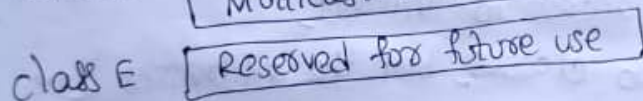
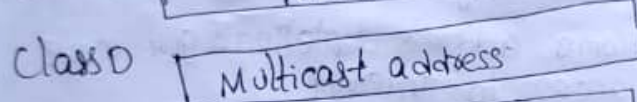
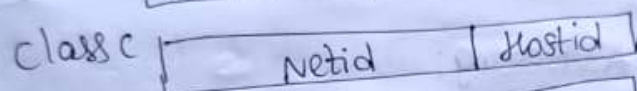
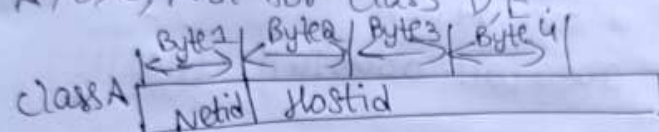
b) Dotted-decimal notation

class A ^{First byte} 0-127 ^{second byte} ^{Third byte} ^{Fourth byte}
class B 128-191
class C 192-223
class D 224-239
class E 240-255

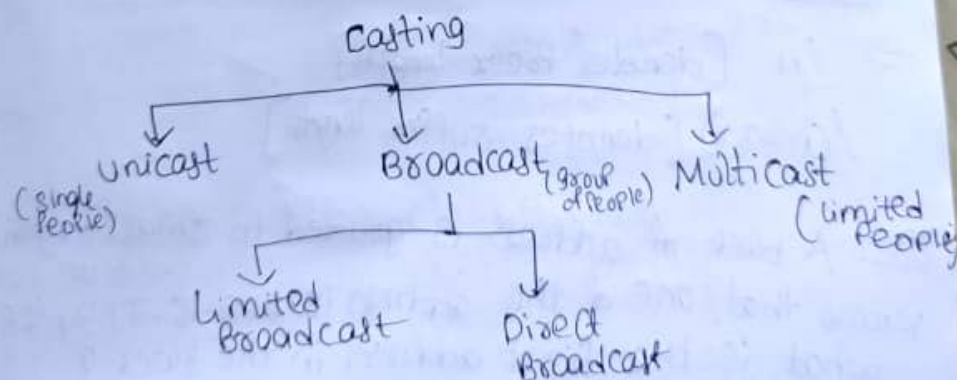
Netid and Hostid :-

Used for subnet mask

→ In classful addressing is applicable for class A, B, C, Not for class D, E



Casting: Transferring Packet to various hosts by using IP address



Mask

Class	8 bits	Binary	Dotted-Decimal
A	11111111	00000000 00000000 00000000	255.0.0.0
B	11111111 11111111	00000000 00000000	255.255.0.0
C	11111111 11111111 11111111	00000000	255.255.255.0

→ Concept doesn't apply for class E

Example:

Find class of each address

00000001 00001011 00001011 11101111

Class A

4) classless Addressing:

→ To overcome address depletion & give more organizations access to Internet.

→ There are no classes, but the addresses are still granted in blocks.

Mask:

→ /n [denotes Prefix length]

/ (n-32) [denotes suffix length].

Ex: A block of address is granted to small organization. we know that one of the address is 205.16.37.39/28. what is the first address in the block?

501: Convert the IP address of 205.16.37.39/28.

2	205
2	102-1
2	51-0
2	25-1
2	12-1
2	6-0
2	3-0
2	1-1
	1

11001101 | 00010000 | 00100101

00100111

If we set 32-28 rightmost bits to 0, we get

11001101 00010000 00100101
00100000

[If they ask first address add 0 value, otherwise last address then 1].

to calculate first address then do AND operation for Address & MASK.

for last address then OR operation.

Internet Protocol 6 (IPv6)

- 128
- 8 groups and each group represents 2 bytes.
- Hex-Digit of 4 bits
- Delimite Uses - colon (:))

→ Give network id & host id of following IPv4 address.

a) 56.89.34.25 - class A (0-127)
56-net id
89.34.25-host id

b) 165.25.242.8 - class B (128-191)
net id - 165.25
host id - 242.8

c) 212.115.65.24 - class C (192-223)
net id - 212.115.65
Host id - 24

d) 227.54.68.26 - class D (224-239)

e) 249.84.69.54 - class E (240-255)

2) If a class B network has a subnet mask 255.255.245.0, what is the maximum number of hosts per subnet.

Sol: Convert subnet mask to binary

255.255.245.0

→ Binary = 11111111.11111111.11110101.00000000

In this 20 bits in the subnet. So $32 - 20 = 12$ taken for host IDs.

Hence total possible values of host IDs are 2^{12}
= 4096. Out of this 4096 values i) Broadcast address ii) subnet network address.

→ The number of address usable for addressing specific host in each network is always $(2^n - 2)$ where n is the number of bits for host ID.

$$\Rightarrow 2^n - 2 \Rightarrow 4096 - 2 = 4094$$

3) Given IP address 211.25.38.26. Find the network ID, 5th host ID, last host ID and broadcast ID of the given network.

Sol: 211.25.38.26, netid - 211.25.38, hostid - 26

Binary representation:

11010011 00011001 00100110

→ The first address in block can be found by setting 32-bits to 0s

00011010

by setting 32-bits

If we see
11 0100

(last address)
If we see

110100

⇒ 5th host
⇒ last host

c) Using
- 8th 10
- host ID
hosts

A) clo
we l

32

Cisco T

* Com
* Dist
* Acco

IP tes

If we set 32-24 rightmost bits to 0, we get
11 010011 00011001 00100110 00000000

(last address)

(08)

211.25.38.0

If we set 32-24 rightmost bits to 1, we get
11 010011 00011001 00100110 11111111

⇒ 5th host id - 211.25.38.4

⇒ last host id - 211.25.38.254

c) Using fixed length subnet divide the given network
196.220.84.32 into two parts, Give first
host ID, last host ID, subnet mask and no. of
hosts connected.

A) class C.

we have 196.220.84 net ID (24 bits)

32 is host id

Cisco Three layer network model

- * Core layer
- * Distribution layer
- * Access layer.

IP tunneling